

02 — Simulation: families, prior, constraints

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1 Generation

One pool of patterns swept over the regime coordinates $(\bar{n}, \tilde{\delta})$: expected count and departure from CSR. Train and test sets are carved from it later. Every structured family gets the same $(\bar{n}, \tilde{\delta})$ distribution, so the regime of a pattern says nothing about which family produced it. Two patterns per parameter θ , so bias and variance can be separated at fixed θ .

Parameterisation

Each θ can be interpreted as three components: (1) the expected count $\bar{n}$; (2) a *shape* - i.e. what length scale the structure lives; (3) *amplitude* - i.e. how strong it is at that scale. Only the amplitude is used to sweep the target $\tilde{\delta}$ at fixed nb .

- **Poisson**: homogeneous, intensity $\bar{n}$. No structure.
- **Thomas**: parents at intensity κ , each with $\text{Poisson}(\mu)$ offspring displaced by $N(0, \sigma^2 I)$. Shape σ , amplitude μ .
- **Nested**: the same construction at two levels — meta-parents at κ , $\text{Poisson}(\mu_1)$ parents at scale σ_1 , $\text{Poisson}(\mu_2)$ children at scale σ_2 . Shape $(\sigma_2, \rho = \sigma_1/\sigma_2, \mu_1)$, amplitude μ_2 .
- **LGCP**: $\Lambda = e^{\mu+Y}$ with $\text{Cov } Y = \sigma^2 e^{-r/s}$. Shape s , amplitude σ^2 .
- **Matérn II**: primaries at λ_p , thinned by a hard core of radius R . No shape; amplitude R .

In each family one further parameter is not free: it is fixed by $\bar{n}$ once the others are chosen ($\kappa = \bar{n}/\mu$ for Thomas, $\kappa = \bar{n}/(\mu_1\mu_2)$ for nested, $\mu = \log \bar{n} - \sigma^2/2$ for the LGCP, and λ_p from $\bar{n} = (1 - e^{-\lambda_p \pi R^2})/(\pi R^2)$ for Matérn). So moving the amplitude at fixed $(\bar{n}, \text{shape})$ keeps the expected count and the scale of the structure unchanged, and changes only its strength. Sending the amplitude to zero gives CSR: for Thomas, $\mu \rightarrow 0$ forces $\kappa = \bar{n}/\mu \rightarrow \infty$, and a superposition of very many parents with about one point each is $\text{Poisson}(\bar{n})$.

$\tilde{\delta}$ increases monotonically with the amplitude (checked on grids), so a target $\tilde{\delta}$ has exactly one amplitude. It is found by root-finding on the closed-form K rather than by simulation.

Validity rules

Four rules remove parameters for which $\tilde{\delta}$ would be a misleading label. The first two restrict the shape, the last two cap the amplitude. Tolerances $\varepsilon = 0.2$, $c = 0.45$; the rules use the window $[r_{\min}(\bar{n}), 0.25]$ with $r_{\min}(\bar{n}) = r_{\min}(\bar{n})$ at 10 expected pairs (`rules.min_pairs` in `configs/simulation`). That floor is a design choice about which scales are worth sweeping and is deliberately fixed: the test's own cutoff (01, $p = 5$) can be refitted without moving the shape prior.

1. **Structure inside $R(\bar{n})$** . $\tilde{\delta}$ only looks at $R(\bar{n})$, so structure sitting mostly below $r_{\min}(\bar{n})$ or above 0.25 would be invisible to it and patterns with the same $\tilde{\delta}$ would not be equally hard. Requiring most of the structure to fall inside the window gives $\sigma \leq 0.098$ (Thomas), $s \leq 0.155$ (LGCP), and matching bounds on the inner and outer nested scales.

2. **Nested levels distinguishable.** Otherwise nested is just a Thomas process. The two scales must be separated ($\rho \geq 2.49$) and both must carry a non-negligible share of the clustering ($1 \leq \mu_1 \leq 4$).
3. **Enough independent units.** $\tilde{\delta}$ is defined at $\bar{n}$, so it only means something if the realised n stays near $\bar{n}$; a few large clusters make n swing. Rule: the coefficient of variation of n , available in closed form, is at most $c = 0.45$.
4. **Matérn packing.** Type II thinning cannot pass a fill of $\pi R^2 \bar{n} = 1$ and degenerates near it; capped at 0.95.

Range of $\tilde{\delta}$

The rules cap how much departure each family can reach. Largest $\tilde{\delta}$ attainable:

$\bar{n}$	100	152	230	348	528	800
Thomas	16.4	26.4	41.6	64.6	99.9	154
Nested	14.1	23.4	38.0	60.9	96.0	149
LGCP	16.9	26.9	41.6	63.1	94.5	147
Matérn II	4.2	5.2	6.4	7.9	9.8	11.8

Matérn is the binding family, and its cap is physical rather than a choice of tolerance (raising the fill from 0.95 to 0.999 moves it by 2%). The range common to all families at all $\bar{n}$ is therefore $\tilde{\delta} \in [0.1, 4]$, inside which $\bar{n}$ and $\tilde{\delta}$ can be drawn independently. The lower end is provisional: any larger floor gives a subset of this pool.

Matérn II is never visibly regular. Write $\mu_R = \frac{1}{2}\bar{n}(\bar{n} - 1)\pi R^2$ for the CSR pairs the hard core forbids ($\mu_R = p$ exactly when $R = r_{\min}(\bar{n})$, so it says whether the hard core is inside the window $\tilde{\delta}$ looks at). As $\lambda_p \rightarrow \infty$ the retained intensity tends to $1/(\pi R^2)$, so the fill $\pi R^2 \bar{n}$ saturates at 1. Two points closer than R are two discs of radius $R/2$ overlapping, so that is a packing fraction of 0.25, against 0.91 for dense packing. Even at its own limit the family is only mildly inhibited, which is why the Matérn row of Fig. 1 looks like CSR at every $\tilde{\delta}$: the cause is the family, not the prior, the $\tilde{\delta}$ cap or any constraint on n . Strong regularity would need a different repulsive model (Matérn III, sequential inhibition, a hard-core Gibbs process or a DPP). The headroom above $\tilde{\delta} = 4$ is also real but uninformative: the feasible maximum is 4.2 at $\bar{n} = 100$ and 11.8 at $\bar{n} = 800$, and every θ there has $\mu_R \geq 47$, where every method already saturates.

Prior

The draw order is part of the definition.

1. $\bar{n} \sim \text{log-uniform}[100, 800]$.
2. $\tilde{\delta} \sim \text{log-uniform}[0.1, 4]$ (Poisson: $\tilde{\delta} = 0$, skip the rest).
3. Shape: each coordinate log-uniform on the box allowed by rules 1–2 at that $\bar{n}$; redraw until valid and until the target $\tilde{\delta}$ is reachable.
4. Amplitude: the unique value hitting the target $\tilde{\delta}$, subject to the rule-3/4 caps.

Because the shape is drawn after $\tilde{\delta}$ and from its own uniforms, the shape distribution is the same at every $\tilde{\delta}$. The amplitude is never drawn, so its distribution is whatever the $\tilde{\delta}$ prior induces; this is the effective prior that any learned estimator will shrink towards.

$\tilde{\delta}$ was drawn under the $p = 10$ tables. The patterns are untouched by the refit to $p = 5$, but $\tilde{\delta}$ is a function of the tables, so every θ carries two numbers: `delta`, the target it was drawn and solved for, and `delta_tilde`, its $\tilde{\delta}$ under the tables now (`scripts/relabel.py`); regimes are binned on the latter. For the cluster families the relabelling is a uniform 1–2% shrink, so their prior is still log-uniform (KS p 0.30–0.58) and still independent of $\bar{n}$ (χ^2 p 0.09–0.90). For Matérn it is nonlinear (median $\times 1.40$), so its prior is no longer log-uniform: the range is 0.126–3.95, and the 8 reporting bins hold 431–2860

θ instead of ≈ 1250 . Re-solving the Matérn amplitudes would restore it, at the cost of re-simulating that family. The amplitude solve itself used those tables, so the tables that generated the sweep are kept frozen at `configs/departure/tables_sweep_p10.npz`; reproducing a stored pattern from its seed needs them, not the current ones.

2 Implemented pipeline

Code

<code>src/cloudforger/simulation/seeding.py</code>	addressed random streams
<code>src/cloudforger/simulation/families.py</code>	rules, shape boxes, $K - \pi r^2$, CV, $\tilde{\delta}$, amplitude solve
<code>src/cloudforger/simulation/processes.py</code>	the five samplers
<code>src/cloudforger/simulation/lgcp_grid.py</code>	LGCP grid size
<code>src/cloudforger/simulation/sweep.py</code>	θ draw, replicate patterns, shards
<code>src/cloudforger/simulation/check.py</code>	mean $\hat{K}$ vs closed form
<code>scripts/simulate.py</code>	CLI: <code>run</code> , <code>merge</code> , <code>check</code>
<code>scripts/feasibility.py</code>	maximum reachable $\tilde{\delta}$
<code>scripts/simulation_figures.py</code>	figures of this document
<code>configs/simulation/config.yaml</code>	$\bar{n}$, $\tilde{\delta}$ range, rules, sizes

Validation

- The closed-form K agrees with the previous implementation to 10^{-10} .
- Every sampler reproduces its own closed-form K : over 500 patterns at each of 8 θ per family, the largest deviation is 2.8 standard errors against a 3.4 threshold, and relative error stays under 3%.
- The draw is exactly log-uniform in the target `delta`. After relabelling to `delta_tilde` (Prior) the cluster families still pass: $\log \tilde{\delta}$ uniform (KS p 0.30–0.58) and independent of $\bar{n}$ (χ^2 p 0.09–0.90), every $(\bar{n}, \tilde{\delta})$ cell within 9% of its expected count. Matérn does not (KS $p < 10^{-3}$; cells -53% to $+121\%$), though $\bar{n}$ and $\tilde{\delta}$ stay independent (χ^2 $p = 0.72$).
- Realised n ranges 44–961, no pattern redrawn. The CV rule never binds in this range (max 0.34 against a cap of 0.45); it would only matter at larger $\tilde{\delta}$.
- The 5% L -test of 01 rejects 4.9% of the 20 000 Poisson patterns.
- The two patterns of a θ agree on rejection 77–87% of the time, so a single pattern is a noisy but not arbitrary read of its regime.

3 Analysis

What the figures show

- **The amplitude does what it should** (Fig. 1): along each row only the amplitude moves, and by $\tilde{\delta} \leq 0.25$ the patterns are visually CSR while the scale of the structure is unchanged.
- **Equal $\tilde{\delta}$ means equal difficulty** for the L test, across families (Fig. 2, left):

$\tilde{\delta}$ bin	(0.1, 0.2]	(0.2, 0.4]	(0.4, 0.8]	(0.8, 1.25]	(1.25, 2]	(2, 4]
Thomas	0.076	0.128	0.336	0.709	0.954	0.998
Nested	0.070	0.131	0.370	0.786	0.974	1.000
LGCP	0.081	0.149	0.387	0.752	0.920	0.987
Matérn II	0.071	0.093	0.290	0.683	0.984	1.000

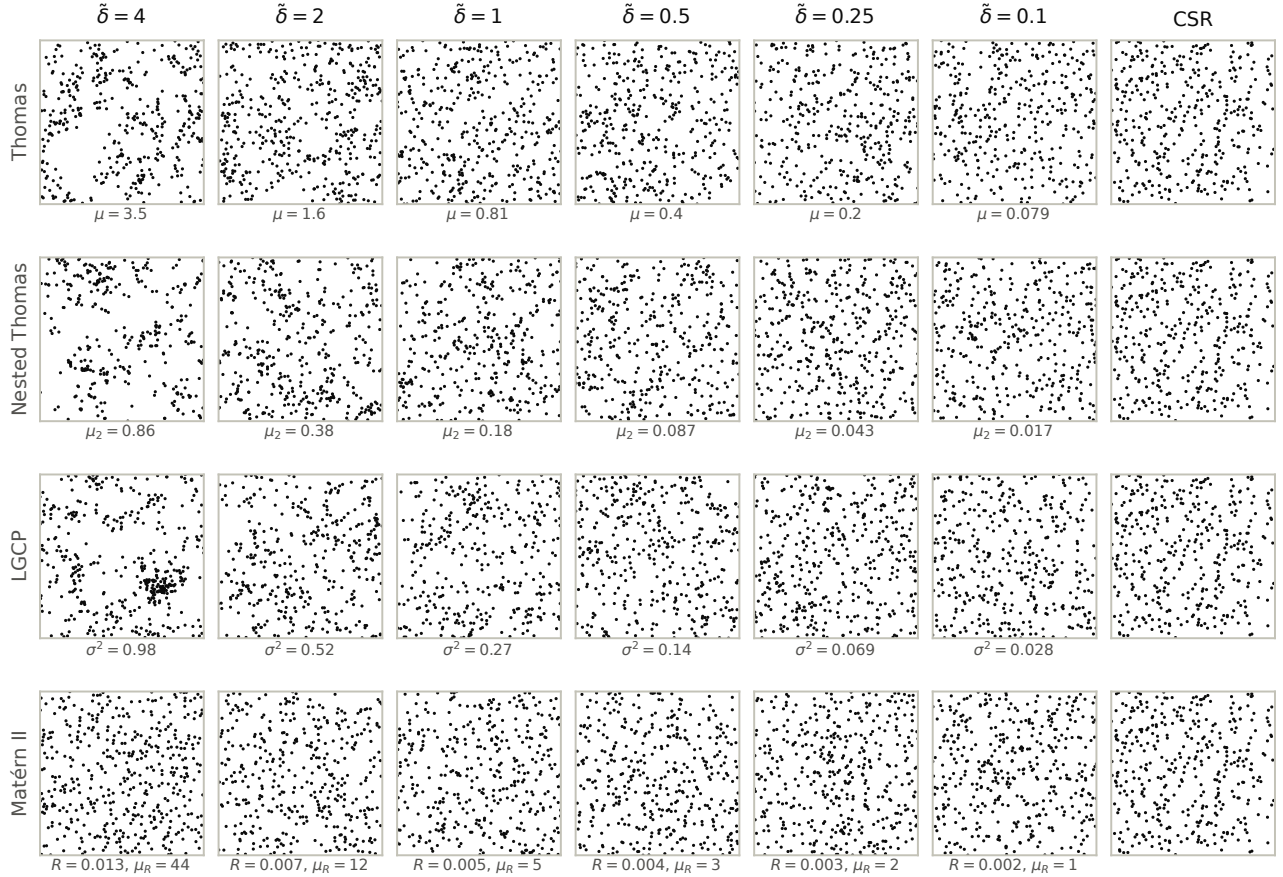


Figure 1: One pattern per cell, $\bar{n} = 400$, fixed shape per family (Thomas $\sigma = 0.03$; nested $\sigma_2 = 0.008$, $\rho = 5$, $\mu_1 = 3$; LGCP $s = 0.04$). Only the amplitude changes along a row, and its value is printed under each panel. For Matérn that is the hard core R , with μ_R , the CSR pairs it forbids: the row looks like CSR throughout because even at $\tilde{\delta} = 4$ the hard core is about a quarter of the mean spacing $1/\sqrt{\bar{n}} = 0.05$.

- $\tilde{\delta}$ **absorbs** $\bar{n}$ (middle): the three $\bar{n}$ terciles agree within 0.03 in every bin, which is what lets the two regime coordinates be read separately.
- **Floor and ceiling** (left and right): below $\tilde{\delta} \approx 0.25$ the classical test is at its noise floor and the median realised S is that of CSR; above $\tilde{\delta} \approx 1.25$ it rejects almost always and S tracks $\tilde{\delta}$. The interesting band is in between.
- **Matérn still lags at the weak end** (left): its hard core is only partly inside $R(\bar{n})$, so part of the departure is seen second hand (01). The cutoff $p = 10 \rightarrow 5$ closed most of the gap — at $\tilde{\delta} \in (0.4, 0.8]$ Matérn went from 0.214 to 0.290 against 0.34–0.39 for the clusters, and it now overtakes them above $\tilde{\delta} \approx 1.25$ — but below $\tilde{\delta} \approx 0.4$ it is still the easier family for a test that looks at nearest-neighbour distances rather than at L .

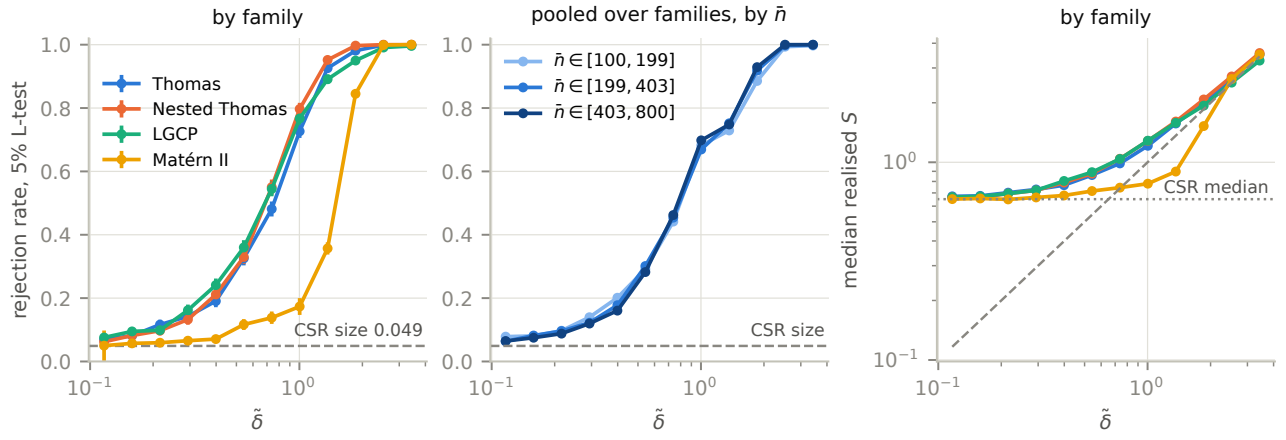


Figure 2: The 5% L-test of 01 on all 100 000 simulated patterns. Left: rejection rate by family, 12 log-bins of $\tilde{\delta}$, 95% intervals. Middle: pooled over families, by $\tilde{n}$ tercile. Right: median realised S (dashed: $S = \tilde{\delta}$; dotted: CSR median).